

Common Probability Distributions

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Outline

- ① Probability Distribution
- ② Some Common Discrete Probability Distributions
- ③ Some Common Continuous Probability Distributions

Probability Distribution

Probability Distribution

- A variable which takes numerical values resulting from random experiments, is called a random variable (RV).
- Recall that probability distribution means the distribution of the probabilities among the different values of a random variable

There are some distributions that often arise frequently in real life. This slide shall discuss a few such distributions

Types of Probability Distribution

Recall that depending on the type of variable, distributions are of two types:

- **Discrete probability distribution:** probability distribution of a discrete random variable
- **Continuous probability distribution:** probability distribution of a continuous random variable

Probability Mass/Density Functions

- The probability distribution of a discrete random variable is given by its probability mass function (PMF)
 - Denoted by $p(x)$
 - $p(x) = P(X = x)$
- The probability distribution of a continuous random variable is given by its probability density function (PDF)
 - Denoted by $f(x)$
 - $\int_a^b f(x) dx = P(a < X < b)$

Only for the continuous case: $P(X = k) = 0$, where k is some constant.

Cumulative Distribution Function (CDF)

- For discrete random variable, $F(x) = P(X \leq x) = \sum_{k=-\infty}^x p(k)$
- For continuous case, $F(x) = P(X \leq x) = P(X < x) = \int_{-\infty}^x f(x) dx$
- It is a non-decreasing function, the largest value $F(x)$ can take is 1

Here, summing/integrating from $-\infty$ means that the lower limit of the sum/integral is the smallest value in the domain of the random variable X .

Some Common Discrete Probability Distributions

Some Common Discrete Probability Distributions

Bernoulli and Binomial Distributions

Bernoulli Process

When a random process or experiment, called a trial, can result in only one of two mutually exclusive outcomes, such as dead or alive, sick or well, full-term or premature, the trial is called a Bernoulli trial.

- Each trial results in one of two possible, mutually exclusive, outcomes. One of the possible outcomes is denoted (arbitrarily) as a success, and the other is denoted a failure
- The probability of a success, denoted by p , remains constant from trial to trial. The probability of a failure, $1 - p$, is denoted by q
- The trials are independent; that is, the outcome of any particular trial is not affected by the outcome of any other trial

Bernoulli Distribution

Let X be the number of successes in a single Bernoulli trial. Then X follows a Bernoulli distribution. It is written as:

$$X \sim \text{Bernoulli}(p).$$

The probability mass function (PMF) of X is given as:

$$p(x) = p^x(1-p)^{1-x}; \quad x = 0, 1$$

- Expected value, $E(X) = 1 \times p + 0 \times (1-p) = p$
- Variance:

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 = (1^2 \times p + 0^2 \times (1-p)) - p^2 \\ &= p - p^2 = p(1-p) \end{aligned}$$

Binomial Distribution

Let X be the number of successes in n Bernoulli trials, where the probability of success is p . Then the random variable X follows a binomial distribution. It is written as:

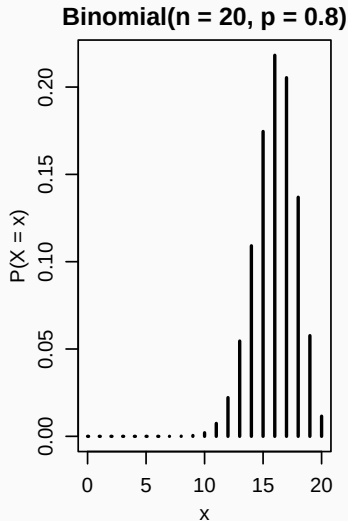
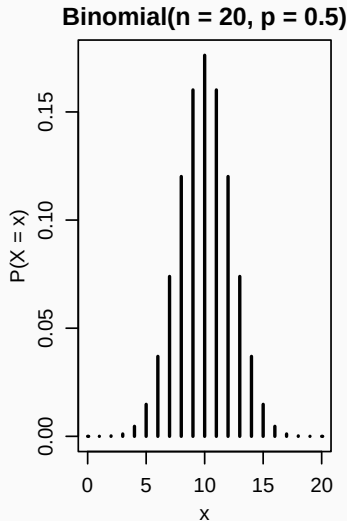
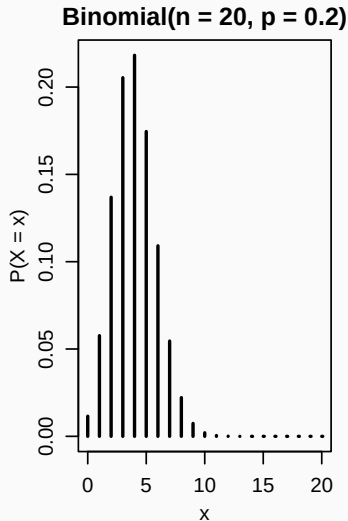
$$X \sim \text{binom}(n, p).$$

The probability mass function (PMF) of X is given as:

$$p(x) = {}^n C_x p^x (1 - p)^{n-x}; \quad x = 0, 1, 2, 3 \dots$$

- $E(X) = np$
- $V(X) = np(1 - p)$

Binomial Distribution: Shape



Examples of Binomial Variables

- Number of defective products among a batch of products
- Number of patients carrying a ceratain pathogen among a group of patients
- Number of heads when a coin is tossed several times

Binomial Distribution: Example

There are 3 multiple choice questions in a MCQ test. Each MCQ consists of four possible choices and only one of them is correct. If an examinee answers those MCQ randomly (without knowing the correct answers).

- ① What is the probability that exactly any two of the answers will be correct?
- ② What is the probability that at least two of the answers will be correct?
- ③ What is the probability that at most two of the answers will be correct?
- ④ What will be the average or expected number of correct answers?
- ⑤ Also, find the standard deviation of number of correct answers.

Binomial Distribution: Example Solution

Each MCQ has 4 options. So, when answering randomly, probability of correct answer, $p = 1/4 = 0.25$. Let the random variable X denote the number of correct answers. Then, $X \sim \text{binomial}(3, 0.25)$.

- ① Probability of two correct answers,
$$P(X = 2) = {}^3C_2 \cdot 0.25^2 (1 - 0.25)^{3-2}$$
- ② Probability of *at least* two correct answers $= P(X = 2) + P(X = 3)$
- ③ Probability of *at most* two correct answers
$$= P(X = 0) + P(X = 1) + P(X = 2)$$
- ④ Expected number of correct answers, $E(X) = 3 \times 0.25 = 0.75$
- ⑤ Variance of number of correct answers, $V(X) = 3 \times 0.25 \times (1 - 0.25)$

The calculations for question 2 and 3, are left as exercises.

Some Common Discrete Probability Distributions

Poisson Distribution

Poisson Distribution

A Poisson random variable is a discrete random variable that counts the number of times a certain event occurs within fixed interval of time, or area, or volume or any other unit.

The Poisson distribution describes the probability of these counts.

Let $X \sim \text{Poisson}(\lambda)$, then the PMF of X is given by:

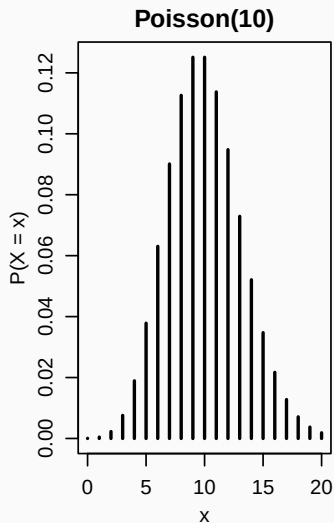
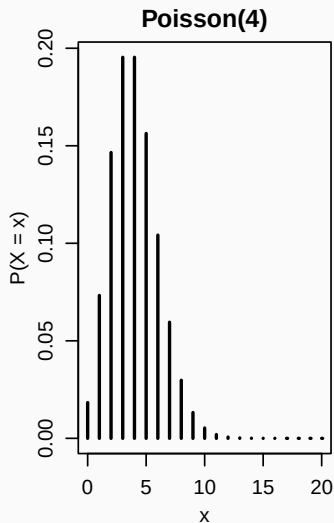
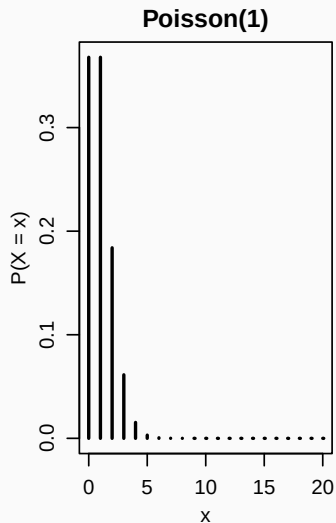
$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; \quad \lambda > 0; \quad x = 0, 1, 2, 3, \dots$$

- λ is the parameter of the distribution
- $E(X) = \lambda$
- $V(X) = \lambda$

Examples of Poisson Random Variables

- Number of phone calls received by a call center in one minute
- Number of typing errors on a page of text
- Number of accidents at an intersection in a day
- Number of emails arriving in an hour

Poisson Distribution: Shape



Assumptions of a Poisson Process

- Events occur independently of each other
- The probability of more than one event in a very small interval is negligible
- The average rate of occurrence λ is constant over the interval
- The probability of an event occurring is proportional to the length of the interval

Poisson Distribution: Example

The average number of errors on a page of a certain magazine is 0.2.
What is the probability that:

- ① a randomly selected page contains 0 (zero) error?
- ② a randomly selected page contains 2 or more errors?
- ③ two consecutive pages have 3 errors?
- ④ What is the average error per page?
- ⑤ Also, find standard deviation of the number of errors in a page.

Poisson Distribution: Example Solution

Here, error rate, $\lambda = 0.2$.

- ① Probability that a randomly selected page contains 0 errors is

$$P(X = 0) = \frac{e^{-0.2} 0.2^0}{0!}$$

- ② Probability that a randomly selected page contains 2 or more errors is

$$\begin{aligned} P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - [P(X = 0) + P(X = 1)] \end{aligned}$$

The above calculations are left as an exercise.

Poisson Distribution: Example Solution (cont.)

- ③ Probability that two consecutive pages have 3 errors:

The error rate for two pages is, $\lambda_{\text{new}} = 0.2 \times 2 = 0.4$ /

Therefore, the desired probability is:

$$\frac{e^{-0.4} 0.4^3}{3!}.$$

- ④ Average error per page is $E(X) = \lambda = 0.2$.
- ⑤ Standard deviation of number of errors in a page,
 $SD(X) = \sqrt{V(X)} = \sqrt{0.2}$.

Some Common Continuous Probability Distributions

Some Common Continuous Probability Distributions

Uniform Distribution

Uniform Distribution

A random variable with sample space $[a, b]$ and equal probability for all intervals of equal length, is called a uniform random variable.

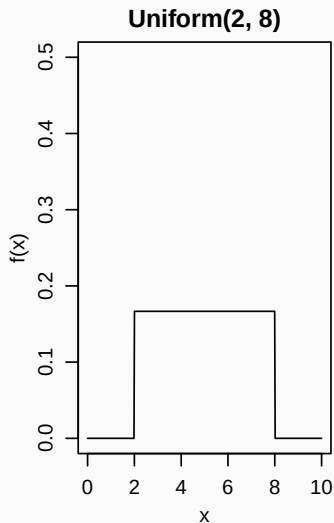
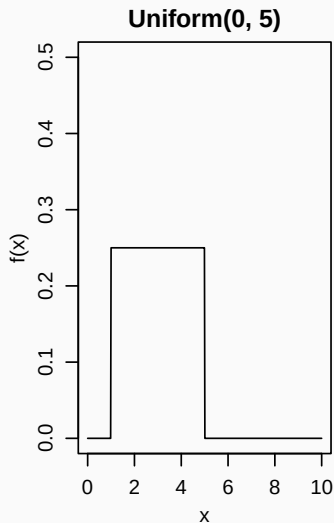
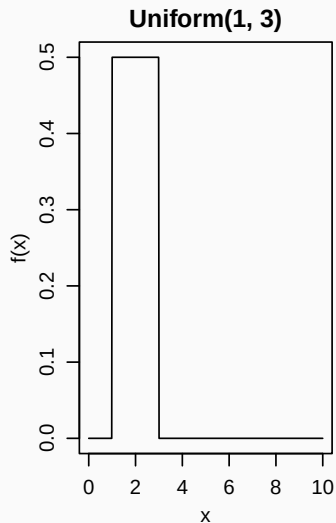
Let X be a uniformly distributed random variable in $[a, b]$, it is written mathematically as $X \sim \text{uniform}(a, b)$.

The PDF of X is given by:

$$f(x) = \frac{1}{b-a}; \quad a \leq x \leq b$$

- $E(X) = \frac{b+a}{2}$
- $V(X) = \frac{(b-a)^2}{12}$

Uniform Distribution: Shape



Uniform Distribution: Example 1

The waiting time (in minutes) for train is $\text{uniform}(10, 50)$. Find the following:

- ① The probability that you have to wait at least 20 minutes
- ② Average waiting time
- ③ Standard deviation of waiting time

Uniform Distribution: Example 1 Solution

Here, wait time, $X \sim \text{uniform}(10, 50)$

- ① Probability of waiting at least 20 minutes:

$$\begin{aligned} P(X > 20) &= P(20 \leq X \leq 50) = \int_{20}^{50} \frac{1}{50 - 10} dx \\ &= \left[\frac{x}{40} \right]_{20}^{50} = 30/40 \end{aligned}$$

- ② Average waiting time, $E(X) = \frac{10+50}{2} = 30$

Question 3 is left as an exercise.

Uniform Distribution: Example 2

Find the CDF of $X \sim \text{uniform}(a, b)$. Then find the CDF the train station waiting example in page 21, i.e., when $a = 10$ and $b = 50$. Using this CDF, find the following probabilities:

- ① Probability that you have to wait less than 25 minutes
- ② Probability that you have to wait more than 25 minutes
- ③ Probability that the wait time is between 26 and 36 minutes
- ④ Probability that the wait time is less than 32 minutes or more than 42 minutes

Uniform Distribution: Example 2 Solution

The CDF is given by:

$$\begin{aligned} F(X) = P(X < x) &= \int_a^x f(x) dx \\ &= \int_a^x \frac{1}{b-a} dx \\ &= \left[\frac{x}{b-a} \right]_a^x \\ &= \frac{x-a}{b-a} \end{aligned}$$

Uniform Distribution: Example 2 Solution (cont.)

From the previous page: $F(x) = \frac{x-a}{b-a}$.

When $a = 10, b = 50$: $F(x) = \frac{x-10}{50-10} = \frac{x-10}{40}$.

$$\textcircled{1} P(X < 25) = F(25) = \frac{25-10}{40} = 15/40$$

$$\textcircled{2} P(X > 25) = 1 - F(25)$$

$$\textcircled{3} P(26 < X < 36) = F(36) - F(26)$$

$$\textcircled{4} P(X < 32 \cup X > 42) = P(X < 32) + P(X > 42) = F(32) + 1 - F(42)$$

Uniform Distribution: Example 3

Derive the expected value of the uniform distribution:

$$\begin{aligned} E(X) &= \int_a^b x \cdot \frac{1}{b-a} dx \\ &= \frac{1}{b-a} \int_a^b x dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b \\ &= \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} \\ &= \frac{b+a}{2} \end{aligned}$$

Some Common Continuous Probability Distributions

Exponential Distribution

Exponential Distribution

Let X be a continuous random variable measuring the waiting time until an event occurs. The event is assumed to occur randomly but at a constant average rate. And let λ be the average number of events in one unit of time.

Then X is said to follow the exponential distribution with rate parameter λ , written as $X \sim \text{exponential}(\lambda)$.

The PDF of X is given by

$$f(x) = \lambda e^{-\lambda x}; \quad x > 0, \quad \lambda > 0.$$

- $E(X) = 1/\lambda$, it denotes the average waiting time between events
- $V(X) = 1/\lambda^2$

Examples of Exponentially Distributed Random Variables

- Waiting time until the next customer arrives at a service counter
- Time between two consecutive phone calls arriving at a call center
- Lifetime of an electronic component assuming a constant failure rate
- Time until a radioactive atom decays
- Waiting time until the next bus arrives when arrivals occur randomly
- Time between failures of a machine operating under steady conditions

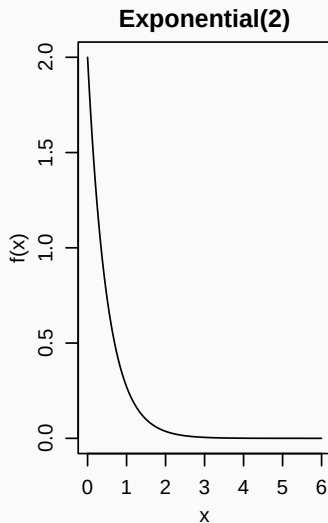
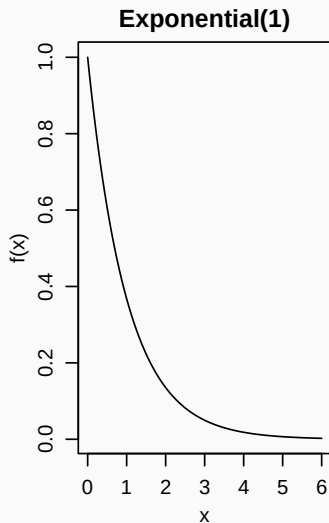
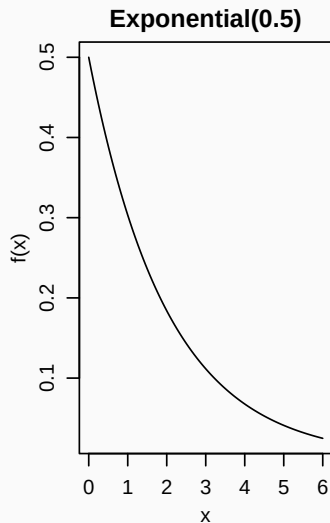
Exponential Distribution and Poisson Distribution

If events occur randomly and independently with but at a constant average rate, λ , then:

- The number of events in a fixed time interval follows a Poisson distribution
- The waiting time between those events follows an exponential distribution

Here, λ is the number of events that occur in one unit of time.

Exponential Distribution Shape



Exponential Distribution: Example 1

Average time required to repair a machine is 0.5 hours. What is the probability that the next repair will take more than 2 hours?

Let, X = time required to repair the machine

Average time required to repara a machine:

$$E(X) = \frac{1}{\lambda} = 0.5 \Rightarrow \lambda = 2$$

Therefore, $X \sim \text{exponential}(2)$

Now, Probability that the next repair will take more than two hours:

$$P(X > 2) = \int_2^{\infty} 2e^{-2x} dx = [-e^{-2x}]_2^{\infty} = [-0 + e^{2 \times 2}] = e^{-4} = 0.0183.$$

Exponential Distribuion: Exercise

Find the CDF of the exponetial distribution.

Some Common Continuous Probability Distributions

Normal Distribution

Normal Distribution

A continuous random variable whose distribution is symmetric and bell shaped around its mean is called a normal random variable.

Let X be a normally distributed random variable with mean μ and variance σ^2 . It is written as

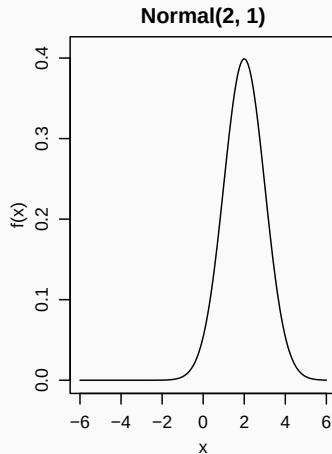
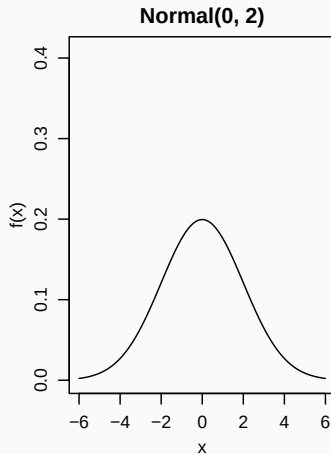
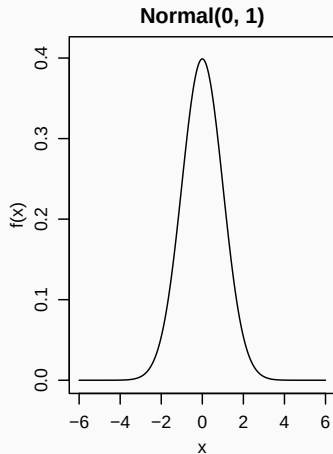
$$X \sim N(\mu, \sigma^2)$$

The probability density function of X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]; \quad -\infty < x < \infty$$

- $E(X) = \mu$
- $V(X) = \sigma^2$

Normal Distribution: Shape



Properties of Normal Distribution

- The distribution is symmetric around the mean μ
- Mean, median and mode are equal
- The spread of the distribution depends on the standard deviation σ
- Larger σ produces a wider and flatter curve
- Total area under the curve is equal to 1

Standard Normal Distribution

A special case of the normal distribution occurs when the mean is 0 and the standard deviation is 1.

If $X \sim N(\mu, \sigma^2)$, then the standardized variable

$$Z = \frac{X - \mu}{\sigma}$$

follows the standard normal distribution.

The standard normal distribution is written as

$$Z \sim N(0, 1)$$

Probabilities for normal variables are usually obtained using standard normal table (Z-table).

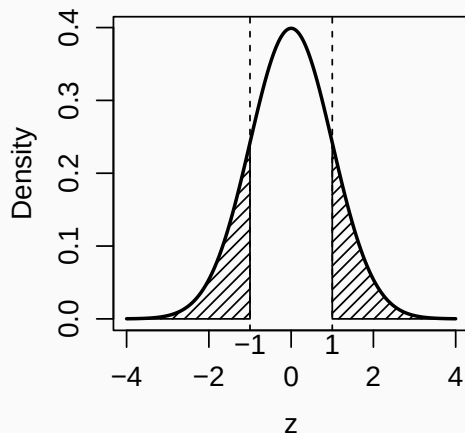
Symmetry of the Standard Normal Distribution

The normal distribution is symmetric around the mean (which is 0 for standard normal distribution). Therefore the following relationship holds:

$$P(Z < -k) = P(Z > k)$$

where k is a positive real number.

The area under the curve of the highlighted regions are equal.



CDF of the Standard Normal Variable

- For a normal random variable $X \sim N(\mu, \sigma^2)$, probabilities are usually calculated using the standard normal distribution CDF
- The cumulative distribution function (CDF) of the standard normal distribution denoted by $\Phi(x)$ is a function that gives the probability: $P(Z < z)$
- The Z-table provides the precalculated cumulative probability: $\Phi(z) = P(Z \leq z)$, which represents the area under the standard normal curve to the **left** of z

Common Probability Intervals

The following types of probability regions commonly appear when working with the normal distribution.

Once the value of Z is obtained, probabilities can be computed using simple rules.

- $P(Z < z) = \Phi(z)$
- $P(Z > z) = 1 - P(Z < z) = 1 - \Phi(z)$
- $P(a < Z < b) = P(Z < b) - P(Z < a) = \Phi(b) - \Phi(a)$
- $P(Z < a \cup Z > b) = P(Z < a) + 1 - P(Z < b) = \Phi(a) + 1 - \Phi(b)$
- $P(Z > k) = P(Z < -k)$, where k is a positive real number

Example: Finding Probability of Normal Random Variable

Let a random variable $X \sim N(10, 25)$. Find $P(X < 4)$ and $P(X > 16)$:

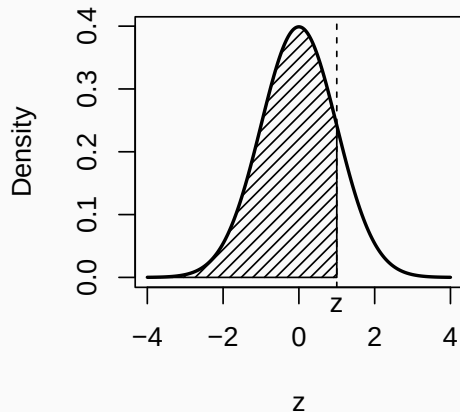
$$P(X < 4) = P\left(\frac{X - 10}{\sqrt{25}} < \frac{4 - 10}{\sqrt{25}}\right) = P(Z < -1.2) = \Phi(-1.2) = 0.115$$

$$\begin{aligned} P(X > 16) &= P\left(\frac{X - 10}{\sqrt{25}} > \frac{16 - 10}{\sqrt{25}}\right) = P(Z > 1.2) = 1 - P(Z < 1.2) \\ &= 1 - \Phi(1.2) \\ &= 1 - 0.885 \\ &= 0.115 \end{aligned}$$

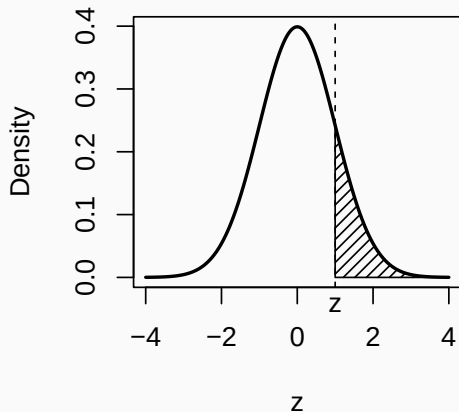
The above values of $\Phi(\cdot)$ are found using the Z-table.

Normal Distribution: Probability Regions

$$P(Z < z)$$

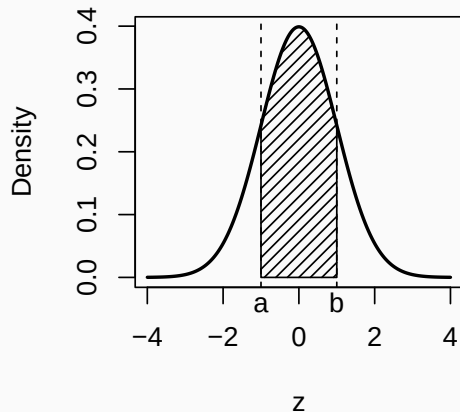


$$P(Z > z)$$

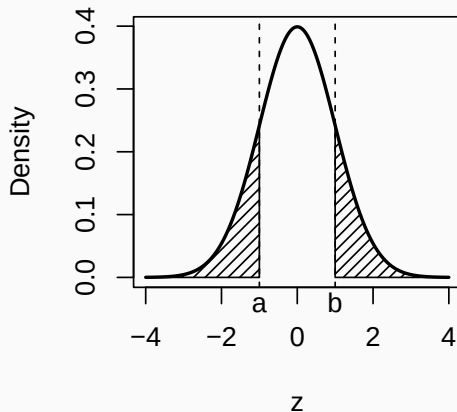


Normal Distribution: Probability Regions (cont.)

$$P(a < Z < b)$$



$$P(Z < a \text{ or } Z > b)$$



Normal Distribution: Example

The heights of students in a university are normally distributed with mean 170 cm and standard deviation 10 cm.

Find the following:

- ① The probability that a randomly selected student is taller than 180 cm
- ② The probability that a student's height lies between 160 cm and 180 cm
- ③ The average height of students

Normal Distribution: Example Solution

Let height $X \sim N(170, 10^2)$

- ① Probability that a student is taller than 180 cm:

$$Z = \frac{180 - 170}{10} = 1$$

$$P(X > 180) = P(Z > 1) = 0.1587$$

- ② Probability that height lies between 160 and 180:

$$Z_1 = \frac{160 - 170}{10} = -1$$

$$Z_2 = \frac{180 - 170}{10} = 1$$

$$P(160 < X < 180) = P(-1 < Z < 1) = 0.6826$$

Thank you.
